KANGYOON LEE

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RESEARCH INTERESTS

Causal Network Traffic Modeling for Preemptive Congestion Control

My research aims to enable preemptive congestion control in datacenter networks. I formalize the causal domain knowledge underlying traffic patterns—temporal user behaviors, physical infrastructure states, and structural feature dependencies—and embed these principles as a priori constraints in predictive models. This yields explainable predictions that operators can trust, enabling proactive intervention before congestion impacts users.

EDUCATION

Aug. 2026 – Present	**Johns Hopkins University** Ph.D. in Computer Science	Baltimore, MD
Mar. 2018 – Feb. 2025	**Hankuk University of Foreign Studies (HUFS)** B.Eng. in Information and Communications Engineering • GPA: **3.96/4.00 (Top 1.8%)** • Thesis: *A Data Forwarding Scheme Using Proportional-Fair Scheduling Algorithm in Edge Network* • Completed 19 months of mandatory military service	Yongin, Korea

WORK EXPERIENCE

Mar. 2025 – May. 2026 — **Research Intern**

Institute: Korea Institute of Science and Technology (Seoul, Republic of Korea)

Role: Network traffic analysis and preprocessing; Multi-modal time-series modeling

- Built multi-modal network traffic DL pipeline for real-time urban crowd density prediction [1][2]

RESEARCH EXPERIENCE

Jun. 2024 – Jun. 2025 — **Extracting Structural Causality in Data Feature Dependencies**

Institute: AI Education Institute, Hansung University (Seoul, Republic of Korea)

- Embedded Bayesian Network-derived feature causality into CTGAN, reducing causal error by 92%

Mar. 2024 – Oct. 2024 — **Modeling Physical Infrastructure Causality for Adaptive Traffic Routing**

Institute: Information Systems Lab, HUFS (Yongin, Republic of Korea)

- Developed adaptive routing algorithm via traffic-infrastructure causal modeling, reducing latency by 9%

Sep. 2023 – Dec. 2023 — **Benchmarking Generative Models for Data Synthesis**

Institute: AI Education Institute, HUFS (Yongin, Republic of Korea)

- Benchmarked generative models for temporal-causal pattern synthesis in multivariate time-series data

Sep. 2022 – May. 2023 — **Uncovering Temporal Causality in Network Traffic Patterns**

Institute: Mobile Distributed Computing Lab, HUFS (Yongin, Republic of Korea)

- Validated temporal causality in traffic patterns through LSTM-based prediction, achieving 94% accuracy vs. 3-9% for memoryless statistical models

PUBLICATIONS

[1] E. Kim, **K. Lee**, Y. Kim, H. Jung, H. Choi, I. Kim, and H. Park. Frequency-Aware Multi-Resolution Architecture for High-Resolution Spatiotemporal Map Reconstruction from Sparse Observations. *38th*

Workshop on Image Processing and Image Understanding (IPIU 2026), 2026, Jeju, Korea (Poster)
[2] Y. Kim, E. Kim, **K. Lee**, H. Park, H. Choi, I. Kim, and H. Jung. Learnable Adjacency Matrix Based Spatio-Temporal Graph Convolutional Network for Traffic Prediction. *Autumn Annual Conference of The Institute of Electronics and Information Engineers (IEIE 2025)*, 2025, Gyeonggi, Korea (Poster)
[3] **K. Lee**, H. Ryou, J. Baek, S. Oh, and I. Doo. A Comparative Study of Synthetic Data with Generative AI Models: Safely Reproducing Sensitive Transportation Worker Data. *International Conference on Advanced Engineering and ICT-Convergence (ICAEIC 2023)*, 2023, Jeju, Korea (Oral **- Best Paper Award)**

PRESENTATION

Korea Police World Expo (KPEX) 2025 — Incheon, Republic of Korea
Presenter, Korea Institute of Science and Technology (KIST)

TEACHING EXPERIENCE

Mar. 2024 – Jun. 2024 — **Network Security Course** at HUFS
Instructor: Prof. Seongho Jeong
- Graded assignments and provided feedback for 30+ students with Q&A sessions

Sep. 2021 – Dec. 2021 — **Algorithm Analysis and Design Course** at HUFS
Instructor: Prof. Sayed C. Shah
- Provided detailed code reviews and debugging support for 50+ students

SCHOLARSHIPS

Mar. 2024	President's Scholarship (**Full tuition**), HUFS
Sep. 2023	Hyoryeong Merit Scholarship, Cheong-gwon-sa Scholarship Foundation
Sep. 2023	Academic Merit Scholarship (**One-quarter tuition**), HUFS
Mar. 2023	Dean's Scholarship (**Half tuition**), HUFS
Sep. 2022	President's Scholarship (**Full tuition**), HUFS

AWARDS

Dec. 2023	Best Paper Award, NCAEIC 2023 – ICT-AES
Dec. 2023	Excellence Award, Capstone Design Project, HUFS
Dec. 2023	Excellence Award, Data Creator Camp, Ministry of Science & ICT
Sep. 2020	Creative Award, National Defense Start-Up Challenge, Top 25/939

NOTABLE ACTIVITIES

- **Community Volunteer** – Regular Donor, Senior & Youth in Crisis, GFoundation, (Mar. 2025 – Present)
- **Conference Staff**, Korea Internet Conference (KRNet) 2023 – Assisted with program coordination, logistics, and technical support
- **Mentor**, AdvICE (Programming Club), HUFS (Sep. 2022 – Dec. 2022) – Guided students in Python algorithm design and competitive programming
- **Communications Specialist**, Republic of Korea Marine Corps (Jun. 2019 – Jan. 2021)

TECHNICAL SKILLS

Programming &Development	Python, C, C++, Java, R, Bash, Git, Linux
Networking & Traffic Analysis	Wireshark, hping3, Ostinato, ns-3, TCP/IP
Machine Learning	PyTorch, TensorFlow, Keras, Numpy, Pandas, Matplotlib
Embedded & IoT	Raspberry Pi, Arduino